Table 12. **Ablation study.** Trained on TinyImageNet with 1-step PGD and evaluated under 100-step PGD, CW, and AutoAttack (AA) with $\epsilon = 1/255$. Results are averaged over 16 datasets. Best and second-best are in **bold** and underline.

Methods	Clean	PGD	CW	AA
CLIP	64.45	0.05	4.06	0.51
$\mathcal{L}_{\text{ce}}$	43.83	29.86	29.25	28.74
$\mathcal{L}_{\text{ce}}$ + KL$(p(x)\|p(x^a))$	45.03	30.12	29.61	29.13
$\mathcal{L}_{\text{ce}}$ + KL$(p(x^a)\|p(x))$	45.05	29.98	29.28	28.80
$\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha)\|\text{Dir}(\alpha_{\text{adv}}))$	36.72	25.01	24.66	24.36
$\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha_{\text{adv}})\|\text{Dir}(\alpha))$	**54.17**	**32.20**	**31.41**	**30.58**

Table 13. **Detailed Ablation study.** Trained on TinyImageNet with 1-step PGD and evaluated under 100-step PGD, CW, and AutoAttack (AA) with $\epsilon = 1/255$. Results are averaged over 16 datasets. Best and second-best are in **bold** and underline.

	Methods	TinyImageNet	Cifar10	Cifar100	STL10	SUN397	Food101	Oxfordpets	Flowers102	DTD	EuroSAT	FGVC Aircraft	ImageNet	Caltech101	Caltech256	StanfordCars	PCAM	Average	H
Clean	CLIP (Radford et al., 2021)	57.96	88.03	60.45	97.03	57.26	83.89	87.41	65.49	40.64	42.66	20.16	59.15	85.32	81.73	52.02	52.08	64.45	
	$\mathcal{L}_{\text{ce}}$	71.24	67.56	38.26	85.89	36.01	28.23	61.30	32.04	24.95	16.13	5.19	32.89	72.16	59.00	20.28	50.11	43.83	
	$\mathcal{L}_{\text{ce}}$ + KL$(p(x)\|p(x^a))$	72.00	70.04	40.68	87.53	37.21	30.39	63.94	33.55	24.84	16.29	4.68	34.52	73.56	60.19	21.30	49.88	45.04	
	$\mathcal{L}_{\text{ce}}$ + KL$(p(x^a)\|p(x))$	70.12	70.62	40.07	87.01	37.88	31.58	63.94	34.01	25.53	14.60	4.92	34.44	74.39	60.64	21.07	49.92	45.05	
	$\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha)\|\text{Dir}(\alpha_{\text{adv}}))$	69.82	63.69	34.74	88.03	27.10	20.78	35.73	10.36	17.93	**23.69**	3.33	15.50	61.47	54.08	11.57	49.83	36.73	
	$\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha_{\text{adv}})\|\text{Dir}(\alpha))$	**74.46**	**81.81**	**54.45**	**91.88**	**41.06**	**53.58**	**74.16**	**47.57**	**31.92**	19.29	**10.95**	**43.20**	**82.39**	**71.53**	**37.32**	**51.20**	**54.17**	
PGD	CLIP (Radford et al., 2021)	0.19	9.57	3.07	23.64	0.62	0.34	0.64	1.62	2.22	0.00	0.00	0.48	5.65	7.19	0.02	0.06	3.46	6.56
	$\mathcal{L}_{\text{ce}}$	**50.96**	39.33	21.64	69.78	20.07	13.50	37.80	19.17	18.30	11.88	2.16	18.47	56.00	42.38	9.33	46.92	29.86	35.52
	$\mathcal{L}_{\text{ce}}$ + KL$(p(x)\|p(x^a))$	50.00	39.19	22.17	70.99	20.07	14.03	39.00	20.10	17.61	12.00	2.25	19.01	57.09	43.33	9.34	45.74	30.12	36.10
	$\mathcal{L}_{\text{ce}}$ + KL$(p(x^a)\|p(x))$	47.64	39.33	21.64	69.69	**20.55**	14.57	38.21	19.97	18.14	11.57	2.07	18.73	57.55	43.01	9.19	**47.80**	29.98	36.00
	$\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha)\|\text{Dir}(\alpha_{\text{adv}}))$	47.66	35.37	18.94	70.18	14.86	10.11	22.13	6.29	13.56	**13.10**	1.41	8.36	47.14	38.52	5.40	47.18	25.01	29.76
	$\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha_{\text{adv}})\|\text{Dir}(\alpha))$	47.56	**43.81**	**25.16**	**73.83**	20.44	**22.86**	**45.11**	**26.79**	**19.47**	2.99	**3.45**	**22.22**	**65.32**	**50.47**	**15.30**	30.37	**32.20**	**40.39**
CW	CLIP (Radford et al., 2021)	0.14	9.91	3.34	26.01	1.16	0.51	0.87	2.03	2.55	0.01	0.00	1.10	6.82	8.17	2.32	0.04	4.06	7.64
	$\mathcal{L}_{\text{ce}}$	**50.16**	38.62	20.76	69.55	18.84	12.46	37.37	18.12	17.23	11.63	2.10	17.70	55.62	41.70	9.23	46.88	29.25	35.08
	$\mathcal{L}_{\text{ce}}$ + KL$(p(x)\|p(x^a))$	49.48	39.02	21.66	70.88	18.95	13.04	38.81	19.22	16.81	11.62	1.74	18.29	56.62	42.77	9.18	45.74	29.61	35.73
	$\mathcal{L}_{\text{ce}}$ + KL$(p(x^a)\|p(x))$	46.54	38.81	20.76	69.28	19.10	13.36	37.67	18.93	17.02	11.40	1.74	17.84	57.02	42.37	9.00	**47.68**	29.28	35.49
	$\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha)\|\text{Dir}(\alpha_{\text{adv}}))$	46.82	35.14	18.31	69.66	14.45	9.71	21.94	6.21	12.82	**12.39**	1.29	8.18	47.15	38.08	5.27	47.16	24.66	29.51
	$\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha_{\text{adv}})\|\text{Dir}(\alpha))$	47.08	**43.30**	**23.92**	**73.55**	**19.20**	**21.68**	**45.38**	**24.95**	**17.87**	2.41	**3.21**	**21.14**	**64.63**	**49.54**	**14.75**	29.89	**31.41**	**39.76**
Auto Attack	CLIP (Radford et al., 2021)	0.00	2.54	1.11	3.18	0.05	0.03	0.03	0.02	0.19	0.17	0.23	0.04	0.10	0.26	0.07	0.12	0.51	1.01
	$\mathcal{L}_{\text{ce}}$	**49.44**	37.87	20.45	69.31	17.41	12.19	36.58	17.81	17.29	11.42	1.86	17.19	54.95	41.19	8.16	46.79	28.74	34.72
	$\mathcal{L}_{\text{ce}}$ + KL$(p(x)\|p(x^a))$	48.48	38.22	20.99	70.66	18.33	12.62	37.99	18.98	16.70	**11.44**	1.71	17.70	56.16	42.29	8.10	45.64	29.13	35.37
	$\mathcal{L}_{\text{ce}}$ + KL$(p(x^a)\|p(x))$	45.80	37.97	20.28	68.95	**18.44**	13.06	37.07	18.77	17.07	11.33	1.56	17.16	56.02	41.76	8.00	**47.52**	28.80	35.13
	$\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha)\|\text{Dir}(\alpha_{\text{adv}}))$	45.88	34.29	17.72	69.58	14.00	9.56	21.72	6.18	12.98	12.05	1.20	8.01	46.76	37.83	4.91	47.07	24.36	29.29
	$\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha_{\text{adv}})\|\text{Dir}(\alpha))$	45.80	**42.32**	**23.03**	**73.15**	18.26	**20.52**	**44.02**	**24.54**	**18.14**	2.26	**2.61**	**20.15**	**63.73**	**48.66**	**12.60**	29.51	**30.58**	**39.09**

Table 14. **Zero-shot adversarial robustness on multi-label dataset MS-COCO (Lin et al., 2014).** All ZSAR models are adversarially trained on TinyImageNet with the FARE (Schlarmann et al., 2024) 10-step PGD setting ($\epsilon = 2/255$), and evaluated under CW-100 at radii $\epsilon \in \{1/255, 2/255, 4/255\}$ plus clean. We report mean Average Precision (mAP), Precision (P), Recall (R), and F1-score (F1) at top-3 predictions. $H(\text{F1@3})$ denotes the harmonic mean of clean and adversarial F1@3. Best and second-best are in **bold** and underline.

	Clean				$\epsilon = 1/255$					$\epsilon = 2/255$					$\epsilon = 4/255$				
Methods	mAP	P@3	R@3	F1@3	mAP	P@3	R@3	F1@3	H(F1@3)	mAP	P@3	R@3	F1@3	H(F1@3)	mAP	P@3	R@3	F1@3	H(F1@3)
CLIP (Radford et al., 2021)	51.96	45.33	46.49	45.9	17.72	25.21	25.85	25.52	32.80	6.67	10.07	10.32	10.19	16.68	3.77	3.84	3.94	3.89	7.17
TeCoA (Mao et al., 2022)	46.34	33.73	34.57	34.14	37.32	30.23	30.99	30.60	32.27	27.94	24.85	25.48	25.16	28.97	**17.63**	19.67	20.17	19.91	25.15
FARE (Schlarmann et al., 2024)	**49.21**	**43.83**	**44.95**	**44.37**	29.18	33.45	34.30	33.86	38.41	16.18	22.44	23.02	22.72	30.05	8.12	13.21	13.55	13.38	20.56
PMG-AFT (Wang et al., 2024)	48.94	41.77	42.83	42.29	29.75	32.32	33.15	32.72	36.89	17.16	23.37	23.98	23.67	30.35	8.78	13.45	13.8	13.62	20.60
TGA-ZSR (Yu et al., 2024)	48.61	37.19	38.13	37.65	**38.23**	32.95	33.79	33.36	35.38	29.03	27.70	28.40	28.04	32.14	15.18	18.64	19.11	18.87	25.14
Comp-TGA (Yu et al., 2026)	44.44	37.62	38.57	38.08	30.82	28.66	29.39	29.01	32.93	25.51	25.48	26.13	25.80	30.76	17.79	20.65	21.18	20.91	27.00
UCAT (Ours)	47.14	41.55	42.62	42.07	37.60	**36.58**	**37.52**	**37.04**	**39.40**	**29.33**	**31.39**	**32.20**	**31.78**	**36.21**	14.61	**21.33**	**21.86**	**21.59**	**28.54**

Table 15. **Zero-shot adversarial robustness on multi-label dataset MS-COCO (Lin et al., 2014).** All ZSAR models are adversarially trained on TinyImageNet with the FARE (Schlarmann et al., 2024) 10-step PGD setting ($\epsilon = 2/255$), and evaluated under **PGD**-100 at radii $\epsilon \in \{1/255, 2/255, 4/255\}$ plus clean. We report mean Average Precision (mAP), Precision (P), Recall (R), and F1-score (F1) at top-3 predictions. $H(\text{F1@3})$ denotes the harmonic mean of clean and adversarial F1@3. Best and second-best are in **bold** and underline.

	Clean				$\epsilon = 1/255$					$\epsilon = 2/255$					$\epsilon = 4/255$				
Methods	mAP	P@3	R@3	F1@3	mAP	P@3	R@3	F1@3	H(F1@3)	mAP	P@3	R@3	F1@3	H(F1@3)	mAP	P@3	R@3	F1@3	H(F1@3)
CLIP (Radford et al., 2021)	51.96	45.33	46.49	45.9	40.21	40.43	41.45	40.93	43.27	24.94	27.85	28.54	28.19	34.93	11.30	15.55	15.94	15.74	23.44
TeCoA (Mao et al., 2022)	46.34	33.73	34.57	34.14	38.90	39.14	40.14	39.63	36.68	42.10	30.54	31.32	30.92	32.45	**31.29**	25.55	26.21	25.87	29.44
FARE (Schlarmann et al., 2024)	**49.21**	**43.83**	**44.95**	**44.37**	43.33	38.63	39.61	39.11	**41.57**	23.82	32.89	33.74	33.30	38.05	11.21	24.31	24.93	24.61	31.66
PMG-AFT (Wang et al., 2024)	48.94	41.77	42.83	42.29	**46.42**	36.00	36.91	36.44	39.15	34.56	32.91	33.75	33.32	37.27	15.51	25.16	25.81	25.47	31.79
TGA-ZSR (Yu et al., 2024)	48.61	37.19	38.13	37.65	44.62	32.33	33.15	32.73	35.02	**42.79**	33.52	34.37	33.93	35.69	29.20	27.31	28.01	27.65	31.88
Comp-TGA (Yu et al., 2026)	44.44	37.62	38.57	38.08	35.26	31.15	31.94	31.53	34.50	33.01	29.63	30.39	30.00	33.56	26.20	26.76	27.45	27.10	31.67
UCAT (Ours)	47.14	41.55	42.62	42.07	44.31	**40.10**	**41.12**	**40.60**	41.32	39.88	**37.47**	**38.43**	**37.94**	**39.90**	28.01	**29.95**	**30.72**	**30.32**	**35.24**

Table 16. **Zero-shot adversarial robustness across 16 single-label datasets.** All methods are fine-tuned on TinyImageNet following TGA-ZSR (Yu et al., 2024), adversarial training uses 2-step PGD (Madry et al., 2017) with $\epsilon = 1/255$. *Average* is the mean across datasets. *H* is the harmonic mean between Clean and the corresponding robust score. Best and second-best are in **bold** and underline.

	Methods	TinyImageNet	Cifar10	Cifar100	STL10	SUN397	Food101	Oxfordpets	Flowers102	DTD	EuroSAT	FGVC Aircraft	ImageNet	Caltech101	Caltech256	StanfordCars	PCAM	Average	H
Clean	CLIP (Radford et al., 2021)	57.96	88.03	60.45	97.03	57.26	83.89	87.41	65.49	40.64	42.66	20.16	59.15	85.32	81.73	52.02	52.08	64.45	
	TeCoA (Mao et al., 2022)	71.24	67.56	38.26	85.89	36.01	28.23	61.30	32.04	24.95	16.13	5.19	32.89	72.16	59.00	20.28	50.11	43.83	
	FARE (Schlarmann et al., 2024)	41.86	79.81	48.27	**94.24**	46.15	**58.90**	**80.98**	47.63	23.09	**24.19**	**15.63**	42.93	78.22	72.05	**43.96**	50.02	53.00	
	PMG-AFT (Wang et al., 2024)	48.60	74.73	43.59	90.41	**51.70**	56.52	79.40	**48.43**	**32.45**	21.76	11.79	**46.74**	**82.49**	**73.59**	41.21	**56.13**	53.72	
	TGA-ZSR (Yu et al., 2024)	**76.60**	79.18	47.37	90.65	43.10	38.90	68.44	39.81	25.69	19.70	8.82	39.27	76.42	66.31	28.44	49.92	49.91	
	Comp-TGA (Yu et al., 2026)	75.00	**81.91**	50.67	90.84	46.45	44.28	72.04	41.41	28.35	23.50	9.12	42.23	78.96	69.25	29.55	49.88	52.09	
	UCAT (Ours)	74.46	81.81	**54.45**	91.88	41.06	53.58	74.16	47.57	31.92	19.29	10.95	43.20	82.39	71.53	37.32	51.20	**54.17**	
PGD	CLIP (Radford et al., 2021)	0.19	9.57	3.07	23.64	0.62	0.34	0.64	1.62	2.22	0.00	0.00	0.48	5.65	7.19	0.02	0.06	3.46	6.56
	TeCoA (Mao et al., 2022)	50.96	39.33	21.64	69.78	20.07	13.50	37.80	19.17	18.30	**11.88**	2.16	18.47	56.00	42.38	9.33	46.92	29.86	35.52
	FARE (Schlarmann et al., 2024)	3.78	7.83	2.80	48.18	5.66	2.45	10.93	6.52	5.75	0.08	0.54	5.20	33.21	20.70	2.31	48.97	12.81	20.63
	PMG-AFT (Wang et al., 2024)	19.18	**51.39**	27.23	72.63	20.05	16.88	44.59	26.43	**20.05**	11.49	3.21	18.09	61.13	43.46	14.80	**55.52**	31.63	39.82
	TGA-ZSR (Yu et al., 2024)	**50.68**	42.16	22.82	72.18	21.57	16.53	39.96	22.44	17.82	11.75	2.88	20.39	58.05	46.18	11.40	48.05	31.55	38.66
	Comp-TGA (Yu et al., 2026)	49.90	40.44	22.43	72.56	**21.61**	16.81	41.56	23.04	18.88	11.66	2.70	20.07	58.80	46.42	10.50	44.98	31.40	39.18
	UCAT (Ours)	47.56	43.81	**25.16**	**73.83**	20.44	**22.86**	**45.11**	**26.79**	19.47	2.99	**3.45**	**22.22**	**65.32**	**50.47**	**15.30**	30.37	**32.20**	**40.39**
CW	CLIP (Radford et al., 2021)	0.14	9.91	3.34	26.01	1.16	0.51	0.87	2.03	2.55	0.01	0.00	1.10	6.82	8.17	2.32	0.04	4.06	7.64
	TeCoA (Mao et al., 2022)	50.16	38.62	20.76	69.55	18.84	12.46	37.37	18.12	17.23	**11.63**	2.10	17.70	55.62	41.70	9.23	46.88	29.25	35.08
	FARE (Schlarmann et al., 2024)	4.10	4.12	2.96	43.35	6.07	3.17	15.15	5.66	4.52	0.12	1.11	5.34	32.50	20.85	4.38	48.86	12.64	20.41
	PMG-AFT (Wang et al., 2024)	13.16	42.10	21.31	65.69	13.12	11.43	28.05	17.53	12.55	8.51	0.99	11.72	52.84	35.68	7.06	14.26	22.25	31.47
	TGA-ZSR (Yu et al., 2024)	**50.80**	42.24	22.64	71.99	20.83	16.03	40.20	21.52	16.97	11.56	2.85	20.01	57.72	45.84	11.23	**48.03**	31.28	38.46
	Comp-TGA (Yu et al., 2026)	49.82	40.40	22.03	72.33	**21.13**	15.95	42.25	22.23	17.55	11.41	2.64	19.94	58.66	46.13	11.09	45.01	31.16	38.99
	UCAT (Ours)	47.08	**43.30**	**23.92**	**73.55**	19.20	**21.68**	**45.38**	**24.95**	**17.87**	2.41	**3.21**	**21.14**	**64.63**	**49.54**	**14.75**	29.89	**31.41**	**39.76**
Auto Attack	CLIP (Radford et al., 2021)	0.00	2.54	1.11	3.18	0.05	0.03	0.03	0.02	0.19	0.17	0.23	0.04	0.10	0.26	0.07	0.12	0.51	1.01
	TeCoA (Mao et al., 2022)	**49.44**	37.87	20.45	69.31	17.41	12.19	36.58	17.81	17.29	**11.42**	1.86	17.19	54.95	41.19	8.16	46.79	28.74	34.72
	FARE (Schlarmann et al., 2024)	0.12	0.03	0.21	10.18	0.84	0.19	0.93	0.60	1.92	0.07	0.06	0.86	10.26	5.59	0.21	5.15	2.33	4.45
	PMG-AFT (Wang et al., 2024)	8.22	41.86	21.18	65.45	7.95	7.34	18.94	12.59	3.13	7.17	0.51	7.90	44.91	28.29	3.22	7.41	17.88	26.83
	TGA-ZSR (Yu et al., 2024)	49.26	40.92	21.75	71.55	**19.88**	15.32	38.84	20.98	17.02	11.26	2.34	19.12	57.11	45.16	9.87	**48.00**	30.52	37.88
	Comp-TGA (Yu et al., 2026)	39.84	30.11	17.23	65.48	15.70	12.50	34.10	18.21	14.84	10.82	1.74	15.87	52.61	40.69	7.38	42.81	26.24	34.90
	UCAT (Ours)	45.80	**42.32**	**23.03**	**73.15**	18.26	**20.52**	**44.02**	**24.54**	**18.14**	2.26	**2.61**	**20.15**	**63.73**	**48.66**	**12.60**	29.51	**30.58**	**39.09**